

Preparations for Demand Response on a Municipal Level

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Abstract—The balance of supply and demand in the electric grid is easier to manage with electricity storages built into the system, in Austria, typically pumped storage hydro power stations are used for this purpose. This article discusses necessary preparations for implementing an alternative to conventional electricity storages, by using private, public, and commercial infrastructure of a rural municipality in Lower Austria. The water supply system, waste water system, thermal-, and pumping processes are prepared to be managed as one virtual energy storage, while maintaining user comfort for inhabitants. Implementing this proposed system is the first attempt to analyze the effectiveness and the user acceptance of automated demand response in Austria. Possible targets for demand side management have been identified, measured, modeled, and equipped with demand response technology. The proposed system aims to show that successful demand response is possible with little investment and without compromising the customer comfort.

I. PROBLEM DESCRIPTION AND RELATED WORK

Seemingly limitless electrical power is one important basis for the affluence of industrial countries today, where the consumption still rises rapidly, decreasing the margin for error. In Europe the power grids provide enough capacity in most cases, but at some usage peaks during the day the grid reserve is under pressure and aids itself with higher market prices during peak loads from the market point of view. Additionally, a continually increasing rate of renewable energy sources, like wind, water, or solar power generation, are integrated into the existing grid, straining it with naturally fluctuating generation [1]. As result of all this, smart grids have become a large field of research all over the world. The goal, to balance the renewable energies with their fluctuating power generation and to smooth the load profiles of customers, leads to a need for a management system on the demand side.

To achieve this goal, it is necessary to identify possible alternatives to conventional energy storages and measure their potential. Unconventional storage systems can be found in every municipality or city. Thermal processes or electrically pumped water towers can be used for storing energy [2]. Also heating, ventilation, air conditioning (HVAC) processes in buildings can be used for storing energy [3]. All these indirect storage processes are called *conceptional energy*

storages or *virtual energy storages*, based on the idea that energy consumption can be postponed or directed by shifting electrical processes to an earlier or later point in time (Fig. 1) [2]. The same concept can be used for heat pumps, water towers, fridges, HVAC, and more, on a municipal level. On such a level, generation and consumption combined with virtual storages, striving for balance within their own domain, can already be called a microgrid. Multiple micro grids connected would result in a smart grid of micro grids. Is one microgrid off balance, other smart grid micro grids in the vicinity can help out. This smart grid is easier to stabilize because storage – virtual or real – is already built in from the beginning at a local level. This is the first step for self-sustaining areas [4]. Self-sustaining means, there is minimal exchange with the electric power grid. This approach can easily be adapted to different rural areas all over the world.

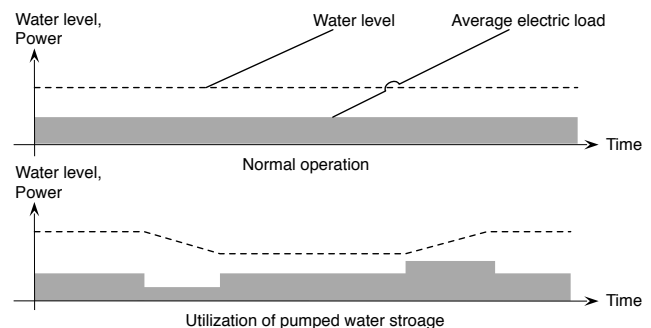


Fig. 1. Utilization of a pumped water storage as conceptual energy storage. The water level is allowed to vary in certain bounds. (modified from [2])

Großschönau with about 1200 inhabitants, a municipality north east of Lower Austria, chose to do this industrial research project with the Vienna University of Technology and the Austrian Institute of Technology (AIT). The community will implement a virtual storage out of existing private, public, and commercial processes, with the goal to simplify future integration of renewable energies. Their interest in renewable energies and communication technology are key. Großschönau is particularly suited for this research, because the area, size, and location is representative or can

be scaled to a multitude of different rural municipalities worldwide.

II. MATERIALS AND METHODS

The first step in a municipality is to identify possible targets to become virtual storages for demand side management (DSM) [5]. Obvious options are HAVC systems and also heat pumps in households, commercial, or buildings. In addition, municipalities often already utilize process control systems for wastewater and freshwater circulation control. Those four kinds of systems were chosen in the Großschönau case, to become a virtual energy storage. Additionally, 2010 7 regional municipalities – one being Großschönau – produced a detailed energy census [6] in which the inhabitants and companies have detailed their annual energy consumption for mobility, heating, and electricity, as well as their eg. annual photovoltaic (PV) generation. The fossil and renewable percentages for each consumed energy mix were recorded as well [7].

A. Locate and quantify potentials

To quantify the total load shifting potential of the municipality, 4 representative kinds of influenceable loads, and 11 solely measured loads were chosen. All systems are monitored and modeled. In a later research phase, these systems will be simulated by system simulation specialists of the AIT, a project partner. All systems are measured and influenced for DSM within a time period of one year. The maximum power consumption of the freshwater and the wastewater system is about 50 kW, HVAC 3 kW, and 2 kW for one exemplary heat pump. These numbers represent the effective potential used for DSM in this research. The overall potential is assumed to be higher and can be shown by upscaling of created models after simulation.

1) *Freshwater*: The freshwater system consist of redundant, alternately operated pumps, tanks, and connecting pipework (Fig. 2). The tanks can be regarded as an energy storage by shifting the pumping process to a different point in time, the electrical energy is stored in the tank and delivered to the grid when the pump is not running. For DSM use, the following tanks, sensors, and pumps were chosen:

- Pumps: Six pumping stations are available, with an overall maximum consumption of 10.5 kW.
- Tanks: Three tanks with volumes of 120 m³, 100 m³, and 80 m³, can be used for the DSM. These tanks can be pre-filled to avoid pumping processes during load peaks or pumped during generation peaks. As shown in Fig. 2 the two upper tanks feed two pressure zones, the *upper* and *lower pressure zone Großschönau*. The third one is a buffer tank for the three wells *Wandl1+2*, and *Agrar*.
- Sensors: Electricity consumption of each pump is measured and in each tank, the water level is measured.

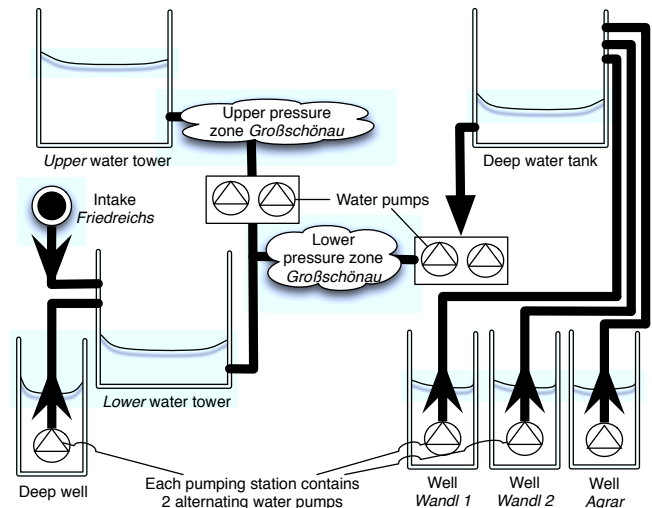


Fig. 2. Schematic drawing of DSM relevant parts in the Großschönau freshwater supply system

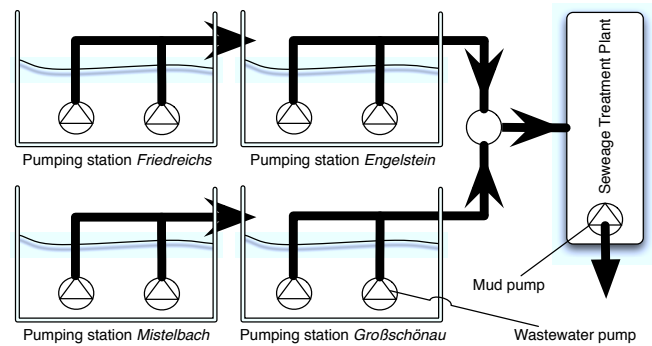


Fig. 3. Schematic drawing of DSM capable parts of the wastewater system in Großschönau

Additionally, some of the mass-flows trough the pipework are measured. All metered data is stored in a database for this research. The consumption of the municipality can be continuously calculated in a created model when combining status of the pumps with levels of tanks and mass-flows.

2) *Wastewater*: The wastewater system consists of a sewage treatment plant and, analogous to the freshwater system, of several delivery pumping stations (Fig. 3). One for every township in the municipality. Because of the sensitive processes of wastewater clarification, the sewage treatment plant itself is not used for DSM. Only one mud pump inside can be used for load shifting, because its task timing is not critical. However, the delivery pumping stations can be seen as electricity storages, just as before described for freshwater tanks.

- Pumps: Four alternating pumps are available for load shifting, with a total consumption of approximately

20 kW total. The mud pump inside the sewage treatment plant consumes 10 kW working.

- Tanks: In contrast to the freshwater system, the wastewater tanks are smaller, hence, offer less virtual energy storage capacity: 4x $\varnothing$ 200 cm and 40-50 cm between min. and max. level.
- Sensors: In each of the tanks, the water level and electricity consumption of the pumps is measured and stored in a database. The amount of wastewater of the municipality can be continuously calculated in the created model when combining status of the pumps with levels of the tanks. The mud pump has no sensors installed, but flexible timing requirements over one day.

3) *HVAC*: In the primary school Großschönau the ventilation system makes sure that the gymnasium is ventilated with fresh air during physical education or public events. The ventilation consumes 1-3 kW, and the non electrically but long-distance heating powered heat exchanger 35 kW. The existing control automation system had no sensors or remote input possibilities. For DSM use, the system needed to be replaced with a new one, adding a CO₂ sensor. The CO₂ concentration in a room can also be seen as a virtual energy storage level. An upper and a lower level of the concentration can be defined and the control system used in the same way as in the wastewater and freshwater tanks. The maximum CO₂ concentration of 600 ppm over the outside concentration is used as maximum level for this research [8].

Next to CO₂ control, remote DSM influencing for this research will be supported, because the HVAC system has even more load shifting potential. The gym has to be ventilated once every day by the morning, so pre-ventilation can additionally be shifted to any time period between midnight and 6 o'clock.

4) *Heat pump*: In the sewage treatment plant, a heat pump is collecting heating power from the rotting process. A recuperator exchanges heat with the control room. As already described in [3], heating processes in buildings are well applicable for DSM because these processes are profoundly inert, simply to control, and measure. For controlling the recuperation process, an additional indoor temperature sensor and a self-developed, influenceable, temperature faking sensor system, communicating with the process control system needs to be installed (described later). The heat pump is exemplary chosen and will be used for upscaling the potential to all pumps in the municipality being used for DSM.

B. Hardware Setup

As mentioned, the fresh and the waste water systems are controlled by a process control system. To enable a single interaction point, the HVAC and heat pump are connected to the process control system as well. The process control

is connected to an object linking and embedding (OLE) for process control (OPC) server [9]. In Großschönau, this server holds only the current data of the system, so that for this research additional communication with a database server is necessary. *MySQL*, an open source relational database management system, based on structured query language (SQL) is used as database server. This server is placed at a remote location in the Vienna University of Technology. Because of security reasons, the database server opens a secure virtual private network (VPN) connection through the internet to connect to the OPC server host. A program written in C# and based on the *OPC Core Components Library* provided by the OPC Foundation, collects all OPC data on-site in Großschönau and transmits it encrypted through the VPN tunnel into the *MySQL* database. To avoid data losses, the program buffers data in case of connection failures and tries to reconnect every 15 s after the connection is lost.

An additional engineering challenge was, that no means of external control were provided for the heat pump by the manufacturer. The pump is designed to be controlled by the outside temperature alone. A lower outside temperature leads to an increase of the inlet temperature of the pump. To trick the heat pump into acting on behalf of DSM, a special controllable temperature sensor system, the pump follows instead, had to be designed. This system allows to simulate a higher or lower outside temperature, so that the load of the pump can be controlled. As a prototype safety measure, the plant operator can turn off the heat pump on a terminal. To provide a control feedback loop, an additional indoor temperature sensor is installed into the control room of the sewage treatment plant.

C. Modeling

For DSM use, the systems has to be modeled and constantly simulated. The systems was modeled in a simulation program by research partners at AIT. After successful modeling of the systems with Großschönau configuration setup from half a year worth of measurements in the database, the simulation model will be optimized by constantly adapting the simulated data with currently measured data. By one day-ahead simulation, the DSM measures possible in this research effort will be predicted, sent to the OPC influencing program developed, and effects constantly measured to provide proof of the effects.

III. RESULTS AND DISCUSSION

So far, more than half a year of measurement data was collected, a simulation model setup, and necessary hardware installed, remotely influenceable by the developed OPC Program waiting for day-ahead simulation input. This section discusses found results, that were essential input to the simulation model configuration and to calculate expected DSM results. Obtained measurements and data from the

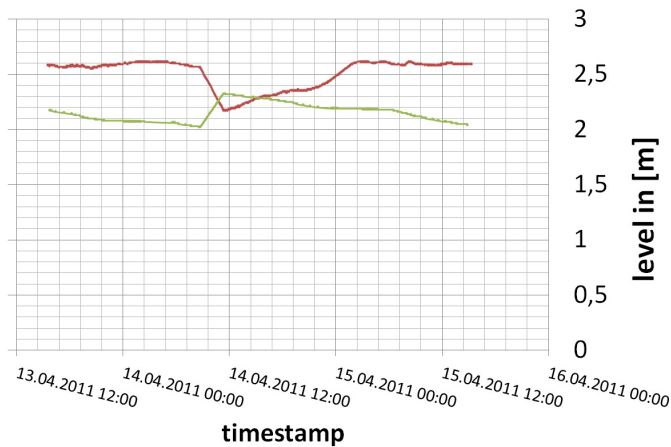


Fig. 4. 2 day measurement of the upper- and lower water tower

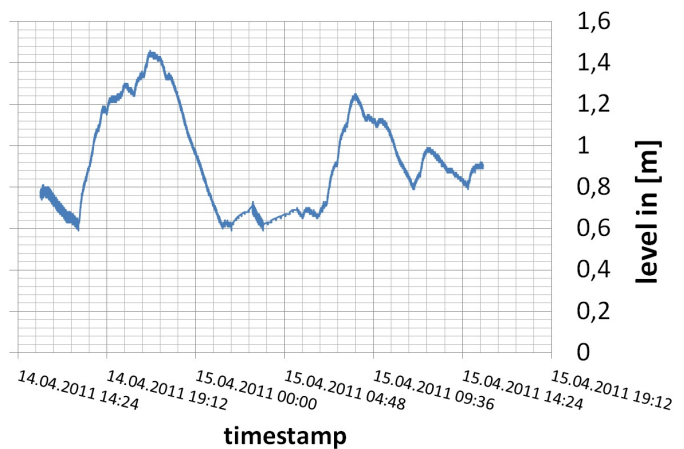


Fig. 6. Measured usage of a wastewater tank during 1 day

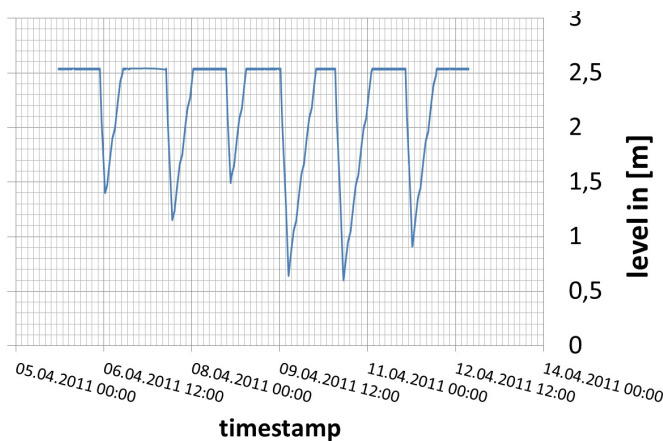


Fig. 5. 10 day measurement of the upper- and lower water tower

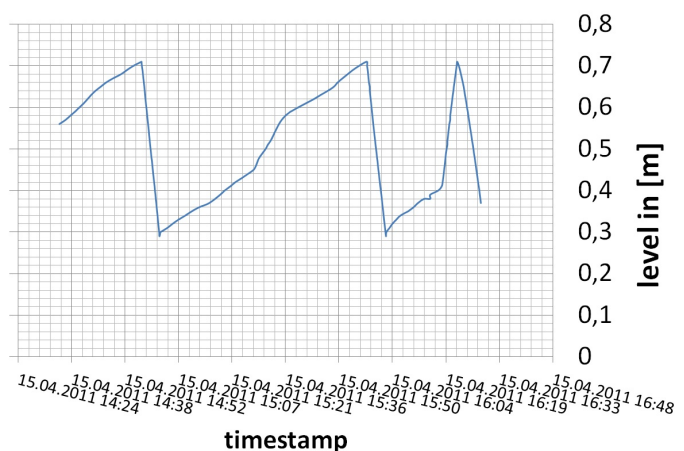


Fig. 7. Measured usage of a wastewater tank during 2 hours

Großschönau energy census were combined to provide expected DSM potentials. A swiss study [10] cites about 150 W energy consumption of auxiliary power for oil and gas heating at a usage of 2000 h/a. That number can be adapted for PV generation, as well as for heat pumps. In the Großschönau energy census there are 387 buildings in the municipality, of which 266 participated. After evaluating the census, it became clear that the buildings are used 93 % over a year. By combining the swiss study with the energy census, a number of approximately 280 kWh per building per year or about 74 MWh for the municipality are available for DSM. This calculation estimate compares to 100 % of the fossil energy that is imported into the municipality (=10% of all imports). With the preliminary results from the measured heat pump this statement can be verified.

As shown in Fig. 4 the two big water towers are inert in operation. So the load shifting potential can be seen as medium, because the time when the pumps are running is not relevant for operation of the system, but the large, energy expensive pumps, operate only once every two days. This results in approximately 1 MWh over one year is available

for load shifting at these pumps (1.5 kW each). It's important to note, that the depicted day is a working day during spring time in this municipality, with only few industrial buildings. The consumption does increase on holidays and in summer, so that the pumps have to run more often.

In Fig. 5 the deep water tank is pictured. The tank is used to buffer three wells and feed both water towers of both pressure zones on a need-to-do basis. The pump runs approximately every $1\frac{1}{8}$ days for 4 hours. In each cycle the pumps consume around 12 kWh, so the consumption over a year is 3.8 MWh. This energy can be used for stabilizing the renewable energy generation through DSM measures. Additionally, with this large time constant, the tank can be pre-filled by the wells or pre-empted for pre-filling the upper tanks to gain even more virtual energy. This DSM technique can also be used to avoid energy intensive pumping sequences during peak-load in the electric grid.

In Fig. 6 one of the four wastewater tanks is measured over the time period of one day. Naturally, the waste water peaks coincide with the electricity grid peaks in Austria. It

is necessary to ensure that the pumps will not run during the peak-load times. A pre-flushing of the tanks will help to avoid these peaks. Fig. 7 shows an other wastewater pump, with a very fast time constant. In this case, load shifting is difficult and hard to predict. Such a pump can still be used for smoothing the fluctuating energy generation from photovoltaic panels.

IV. CONCLUSION AND OUTLOOK

Necessary electricity storages for electric grid stabilization or renewable energies integration can be implemented through existing thermal, pumping, and HVAC processes already present in municipalities. The exact amount of energy loads that can be shifted needs to be determined on a case-to-case basis, but 10% of all electricity imports of a municipality seem to be achievable.

With measurement results and census information, the load shifting potential for the whole municipality can be calculated and the research project can start influencing loads according to simulation models configured and fed by measurement data. The communication program between the Vienna University of Technology and Großschönau can influence the process control system through the local OPC server. The pump control system is equipped with additional software levels for the tanks. The new CO₂ sensors for HVAC influencing at the gymnasium is also connected to the control system and equipped with new CO₂ software levels. The simulation predictions developed at the AIT will take over control through a OPC influencing program, for continued events of several hours in this one year long research effort, to provide measurement data on effects of different taken DSM measures. This experimental phase will be monitored closely by a members of the Vienna University of Technology, AIT, and Großschönau, to avoid drawbacks in consumer comfort or even physical damage for equipment or inhabitants.

ACKNOWLEDGMENT

The authors would like to thank Großschönau for their outstanding initiative and cooperation in this ongoing industrial research effort to make the world a better, more efficient place. This research and publication was made possible by the Federal Government of Austria – managed by the Austrian Research Promotion Agency and the Vienna University of Technology. This paper is based on the findings of the project titled *Großschönau als virtueller Energiespeicher* and was granted 80% of all costs as industrial research in the *Neue Energien 2020 3rd Call* by the Climate and Energy Fund in the topic area *Energy systems, networks and users*.

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